1-to-1 Correspondence: The Concrete Step Counter

Numbers to 5 & 10 • Standard: CCSS.MATH.K.CC.B.4

1. MATHEMATICAL CONCEPT & STACK APPROACH: Establish one-to-one correspondence between physical digit blocks and step tokens on the Place Value Tray and calculator screen increments.

Counting by ones is repeated addition of the unit 1 (n → n + 1). In a stack-based approach, the current total waits on Shelf Y while the unit step (1) enters Workbench X; pressing [+] combines them into the new total.

2. MODELING WITH BLOCKS (WHEN APPLICABLE) & STUDENT PREDICTION

Published Block Kit: Place Value Studio (Digit Blocks 0–9 & Gold [+1] Step Tokens)

Hands-On Setup: Place the [1] Digit Block into the Ones socket of the Place Value Tray. Keep Gold [+1] step tokens ready beside the tray.

Student Prediction (Before Calculator): When we place 1 Gold [+1] step token and step the Ones block up by 1, what digit block replaces [1]? (Prediction: [2])

3. CALCULATOR EXECUTION (SC-6 KEYCAPS & 4-LEVEL STACK TRACE)

Keystroke Flow: [1] [ENTER] [1] [+] [1] [+]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[1]	1	First count placed: 1 enters register X
[ENTER]	1	1 copied to register Y: holds the current count
[1]	1	Gold [+1] step token added: 1 enters register X
[+]	2	Press [+]: 1 + 1 = 2! Swap Ones block to [2]
[1]	1	Next Gold [+1] step token added: 1 enters X
[+]	3	Press [+]: 2 + 1 = 3! Tray [3] and screen match perfectly

4. STORY CONTEXT (How Ten Fingers & an Empty Shell Created Place Value & The 1-to-10 Staircase)

Long before paper or keyboards existed, ancient shepherds counted flocks with clay pebbles ('calculi' in Latin—the root of 'calculator'!) and Maya astronomers drew a shell for zero so an empty socket had its own symbol. Stepping 1 token at a time on the Place Value Tray while pressing [1] [+] builds the exact 1-to-10 Staircase from Chapter 3 of From Blocks to the Stack.

Core Investigation (Student Warm-Up A):

Place the [4] Digit Block in the Ones column of your Place Value Tray. Add 1 Gold [+1] step token. Write the keystrokes to show 4 + 1 on the calculator.

More, Less & Zero: Finding Differences

Comparing Sets & Magnitude • Standard: CCSS.MATH.K.CC.C.6

1. EXTENSION CONCEPT (WARM-UP B), BLOCKS & PREDICTION

Concept & Stack Approach: Comparison subtraction finds the difference ('how many more?') between two quantities. In stack order, the larger reference quantity sits on Waiting Shelf Y while the smaller quantity enters Workbench X before pressing [-].

Blocks & Prediction: Place the [5] Digit Block in Row 1 (top row) and the [2] Digit Block in Row 0 (bottom row) of the Ones column on the Place Value Tray. Prediction: Look at the two rows on the tray. Which row has the larger quantity? How many more is 5 than 2?

2. EXTENSION CALCULATOR SEQUENCE & STACK TRACE

5 ENTER 2 -

Key Pressed	Screen (X)	Internal 4-Level Stack Effect
5	5	Row 1 quantity (5) enters register X
ENTER	5	5 lifted to register Y (first set)
2	2	Row 0 quantity (2) enters register X
-	3	Minus subtracts X from Y: Row 1 has 3 more!

3. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Reciting number names faster than placing physical step tokens (losing 1-to-1 correspondence).

Hands-On Block Remediation: Have the student place 1 Gold [+1] token on the Place Value Tray and step the Ones Digit Block first, say the number aloud, and only then press [1] [+] so the tray and Register X stay locked in step.

Misconception 2: Confusing the action of adding 1 more with the total quantity on the tray (cardinality).

Hands-On Block Remediation: Ask the student to point to the Digit Block in the Ones socket of the Place Value Tray and verify that its numeral matches Register X on the screen.

4. EXTENSION MATHEMATICAL INSIGHT

Comparison subtraction finds the difference ('how many more?') between two quantities. In stack order, the larger reference quantity sits on Waiting Shelf Y while the smaller quantity enters Workbench X before pressing [-].

Counting, Cardinality & The One-More Step Counter

Name: _____ Date: _____ Class: _____

STUDENT REFERENCE GUIDE: HOW COUNTING MAPS TO THE CALCULATOR

- Every Gold [+1] step token added on the Place Value Tray matches one count on the calculator screen.
- A [0] block in the Ones socket represents zero [0]. Press [C] to start a fresh count.
- To step count by 1: press [1] [ENTER] [1] [+], then press [1] [+] as you step your Ones digit block from [1] to [2], [3], [4]...

Key Sequence: `1` `ENTER` `1` `+` `1` `+`

Investigation 1: Place [3] in Row 1 and [2] in Row 0 of the Ones column. Write the keystrokes to find the total.

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Shelves:

Top Shelf (Line Y):

Bottom Shelf (Line X):

Combine (+ or −):

3. My Prediction: _____

Final Verified Answer: _____

[ ] Checked on SC-6

Investigation 2: The Ones column holds [5]. If you subtract [2], what digit block remains?

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Shelves:

Top Shelf (Line Y):

Bottom Shelf (Line X):

Combine (+ or −):

3. My Prediction: _____

Final Verified Answer: _____

[ ] Checked on SC-6

Investigation 3: Clear the calculator by pressing [C]. What Digit Block represents an empty Ones count?

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Shelves:

Top Shelf (Line Y):

Bottom Shelf (Line X):

Combine (+ or −):

3. My Prediction: _____

Final Verified Answer: _____

[ ] Checked on SC-6

Investigation 4: Step count by ones starting at 1: [1] [ENTER] [1] [+]. Press [1] [+] two more times. What is X?

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Shelves:

Top Shelf (Line Y):

Bottom Shelf (Line X):

Combine (+ or −):

3. My Prediction: _____

Final Verified Answer: _____

[ ] Checked on SC-6

Unit 01: Counting, Cardinality & The One-More Step Counter — Answer Key

Complete solutions and SC-6 verification keystrokes for Pre-K – Kindergarten

1. CORE (WARM-UP A) & EXTENSION (WARM-UP B) SOLUTIONS

Core Investigation (Warm-Up A): Place the [4] Digit Block in the Ones column of your Place Value Tray. Add 1 Gold [+1] step token. Write the keystrokes to show 4 + 1 on the calculator.

Core Solution: 5 ([5] Digit Block). Keystrokes: [4] [ENTER] [1] [+]

Core Check Keys: 4 ENTER 1 +

Extension Challenge (Warm-Up B): Row 1 holds [6] and Row 0 holds [2] on your Place Value Tray. How many more is in Row 1? Write the keystrokes.

Extension Solution: 4 more. Keystrokes: [6] [ENTER] [2] [-]

Ext. Check Keys: 6 ENTER 2 -

2. STUDENT LAB WORKSHEET SOLUTIONS (INVESTIGATIONS 1–4)

Investigation 1: Place [3] in Row 1 and [2] in Row 0 of the Ones column. Write the keystrokes to find the total.

Expected Answer: 5

SC-6 Verification Keys: 3 ENTER 2 +

Investigation 2: The Ones column holds [5]. If you subtract [2], what digit block remains?

Expected Answer: 3

SC-6 Verification Keys: 5 ENTER 2 -

Investigation 3: Clear the calculator by pressing [C]. What Digit Block represents an empty Ones count?

Expected Answer: 0 (Zero)

SC-6 Verification Keys: C

Investigation 4: Step count by ones starting at 1: [1] [ENTER] [1] [+]. Press [1] [+] two more times. What is X?

Expected Answer: 4

SC-6 Verification Keys: 1 + 1 +

Hands-On Block & Stack Verification Protocol

Students first build the physical block model and write their mathematical strategy and prediction before verifying their result on the StackCalc calculator.